Brian Xu

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SUMMARY

- Seeking Software Engineering internship.
- Interests include regression analysis, optimization, stochastic processes, data viz, and sports analytics.
- 3-time Texas Regional Team volleyball athlete.

EDUCATION

- **Rice University**
  B.S. Computer Science & Math, GPA: 3.87/4.0
  August 2020 - May 2024

EXPERIENCE

- **Chemstations**
  DevOps Intern
  Eliminated time spent finding code errors by developing a continuous integration and continuous delivery pipeline for ChemCAD software with Jenkins and Subversion, enabling automatic compilation and testing of code when committing to Source Code Manager.
  December 2019 to January 2020

- **RiceApps**
  Mobile App Developer
  Implemented calendar integration for upcoming art events and fixed UI bugs for the Rice Public Art App.
  June 2021 to August 2021

- **Research**
  Undergraduate Researcher
  Surveying latest techniques and methods for symbolic regression under supervision of Dr. Meng Li.
  September 2021 to Present

- **COMP 140**
  Teaching Assistant
  Assisting students with assignments and understanding of course material during office hours as well as grading assignments and exams.
  August 2021 to Present

AWARDS & RECOGNITION

- **National Merit Scholarship Recipient**
  2020

SKILLS & COURSEWORK

- **Languages**
  Python, R, Java, JavaScript, SQL

- **Tools & Technologies**
  HTML5/CSS, React/React Native, LaTeX, Git/GitHub, Hadoop, AWS Elastic MapReduce, Spark, Excel, dplyr, ggplot2, NumPy, pandas

- **Skills**
  Problem Solving, Data Analysis, Data Wrangling, Data Visualization, Program Design

- **Coursework**
  Computational Thinking, Data Structures, Algorithms, Program Design, Tools and Models for Data Science, Discrete Math, Elements of Analysis, Linear Algebra, Calculus I, II, & III, Probability

PROJECTS

- **Volleyball Points Per Pass** [GitHub]
  Created a less subjective and more nuanced volleyball passing metric that grades passes based on their win contribution. Designed and implemented a Markov chain model to do so. Submitting paper to MIT Sloan Sports Analytics Conference Paper Competition.
  R

- **Volleyball Linear Regression** [Poster]
  Finalist in the 2020 MIT Sloan Sports Analytics Conference Poster Competition. Designed and implemented a linear regression model that that explains 95% of the variation in win percentage for NCAA Division 1 Men's Volleyball Teams.
  R

- **Graph Traversal**
  Implemented BFS, Recursive DFS, Iterative DFS, and Dijkstra's algorithm to find routes between locations on the Rice campus.
  Python

HOBBIES & INTERESTS

Volleyball · Esports · Statistics · Math